(Paper Rough Draft) Tutorial: Comparing Permutativity G-Pseudoalgebra and $\mathscr{F}_G$-Spaces

Zih-Yu Hsieh

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Contents

Here I tried to contain all possible details / definitions. I'll see which ones are supposed to be cut eventually. For some of the stuffs that can be omitted, check the SymMonCat paper for reference.

1 Abstract (ND)

Symmetric monoidal categories are central in modern algebraic topology, especially algebraic K-theory relying on constructing spectra out of symmetric monoidal categories. It is known that the two inputs - the categories with action by an operad $\mathscr{P}$ in the category **Cat** of small categories, or suitable categories described by based finite sets $\mathscr{F}$ - are equivalent description for symmetric monoidal categories.

Within equivariant K-theory, similar machineries have been established, where the input for constructing G-spectra are described by an operad $\mathscr{P}_G$ in the category $\mathbf{Cat}_G$ of small G-categories, or categories described by based finite G-sets $\mathscr{F}_G$. The equivalence of the two within equivariant case is a natural question to ask, and ponders the question of "what is symmetric monoidal G-categories".

We'll provide an exposition on the two inputs, and establish a 2-functor $\mathbb{R}_G$ that generates $\mathscr{F}_G$-pseudoalgebra from $\mathscr{P}_G$-pseudoalgebra, relating the two.

2 Introduction (ND)

Fix G as a finite group. Let P_G be the *associativity G-operad* in $\mathbf{Set}_G$, defined by $P_G(n) := \mathbf{Set}(G, \Sigma_n)$, with structure maps defined by post composition

$$\gamma : \mathbf{Set}(G, \Sigma_k) \times \prod_{r=1}^{k} \mathbf{Set}(G, \Sigma_{j_r}) \cong \mathbf{Set}\left(G, \Sigma_k \times \prod_{r=1}^{k} \Sigma_{j_r}\right) \xrightarrow{\gamma \circ -} \mathbf{Set}(G, \Sigma_j),$$

, where $j = \sum_{r=1}^{k} j_r$, and $\gamma : \Sigma_k \times \prod_{r=1}^{k} \Sigma_{j_r} \to \Sigma_j$ is the structure map for associativity operad $\mathscr{M}$ in **Set**. This is naturally a free $(G \times \Sigma)$-operad, as the right $(G \times \Sigma_n)$-action at each $P_G(n)$ is free.

However, it is mentioned in [1] that such G-operad is not finitely generated, as a rough explanation, for every $n \in \mathbb{N}$, there exists $m > n$ and a function $f : G \to \Sigma_m$, such that it cannot be generated using $\{P_G(k)\}_{k \leq n}$ with the operadic structure map and the $(G \times \Sigma)$-actions.

As a substitution, [1] defines a finitely-generated suboperad $Q_{G,\mathfrak{D}} \hookrightarrow P_G$ (for some parameter $\mathfrak{D}$ that's associated with choice of transitive G-sets), such that after chaotic generalization the two are equivalent in $\mathbf{Cat}_G$, and having equivalent categories of operadic algebras.

We'll mimic the statement, define an intermediate operad $Q_{G,\mathfrak{D}} \hookrightarrow Q_G \hookrightarrow P_G$, promote it to $\mathscr{P}_G$ in $\mathbf{Cat}_G$, and *attempt to* modify the proof in [9] to show the equivalence of 2-categories between $\mathscr{P}_G$-**PsAlg** and $\mathscr{F}_{G,\mathbb{R}}$-**PsAlg**.

Future task (assume this works): Prove the equivalence of operads/algebras/pseudoalgebras with the original permuativity G-operad.

Current Goal: Add in extra axioms, so that there exist two 2-functors

$$\mathbb{R}_G : \mathscr{P}_G\text{-}\mathbf{PsAlg} \leftrightarrows \mathscr{F}_{G,\mathbb{R},\_}\text{-}\mathbf{PsAlg} : \mathbb{Q}_G$$

inducing an isomorphism of (strict) 2-categories. Here, $\_$ is some (yet to be determined) axiom.

3 Categorical Preliminaries

3.1 2-Categories

Definition 3.1. A category $\mathscr{C}$ is a *2-category*, if for each object $X, Y \in \mathscr{C}$ (called *0-cells*), the homset $\mathscr{C}(X,Y)$ is itself a category (called the *hom-category*). Also, for every 0-cells $X, Y, Z \in \mathscr{C}$, the *composition* $\circ : \mathscr{C}(Y,Z) \times \mathscr{C}(X,Y) \to \mathscr{C}(X,Z)$ is a functor.

Every morphism $f \in \mathscr{C}(X,Y)$ is called a *1-cell*, while given any 1-cells $f, g \in \mathscr{C}(X,Y)$, a morphism $\alpha : f \Rightarrow g$ in the hom-category $\mathscr{C}(X,Y)$ is called a *2-cell*.

Here, $\circ$ must be strictly unital and associative.

Remark 3.2. There is a more general notion of a *bicategory*, which relaxes the composition functors to be associative and unital up to coherent natural isomorphisms, and in some literature this is called 2-categories. Here we'll take the strict associativity and unitality as axioms for 2-categories. For more general bicategory, cf. [7].

Example 3.3. Let **Cat** denote the *category of small categories*, where the objects are small categories, the hom-categories $\mathbf{Cat}(\mathscr{A}, \mathscr{B})$ are functors between small categories $\mathscr{A}$ and $\mathscr{B}$, and the morphisms in $\mathbf{Cat}(\mathscr{A}, \mathscr{B})$ are natural transformations between functors.

This specifies **Cat** to be a 2-category. Also, a precise description of the horizontal composition of 2-cells is as follow: Given functors $F_1, F_2 \in \mathbf{Cat}(\mathscr{A}, \mathscr{B})$ and $G_1, G_2 \in \mathbf{Cat}(\mathscr{B}, \mathscr{C})$, natural transformations $\mu : F_1 \Rightarrow F_2$ in $\mathbf{Cat}(\mathscr{A}, \mathscr{B})$ and $\nu : G_1 \Rightarrow G_2$ in $\mathbf{Cat}(\mathscr{B}, \mathscr{C})$, then $\nu * \mu := \nu \circ F\mu$ defines a horizontal composition.

3.2 Chaotic-Forgetful Adjunction

Given any small category $\mathscr{A} \in \mathbf{Cat}$, its object $\mathrm{Ob}(\mathscr{A})$ forms a set; moreover, any functor $F : \mathscr{A} \to \mathscr{B}$ also has an underlying map of objects $F : \mathrm{Ob}(\mathscr{A}) \to \mathrm{Ob}(\mathscr{B})$. Conversely, given a set, it has certain category structure:

Definition 3.4. Given a set X, define $\mathscr{E}X$ to be the *chaotic category* with underlying object set $\mathrm{Ob}(\mathscr{E}X) := X$, by encoding a unique isomorphism $x \xrightarrow{\sim} y$ between any element $x, y \in X$. The composition

$$(x \xrightarrow{\sim} y \xrightarrow{\sim} z) = (x \xrightarrow{\sim} z) \tag{3.5}$$

using the uniqueness.

Given a set map $f : X \to Y$, define the functor $\mathscr{E}f : \mathscr{E}X \to \mathscr{E}Y$ so that the underlying object map is f, and for each morphism $x \xrightarrow{\sim} y$ in $\mathscr{E}X$, $\mathscr{E}f(x \to y) := f(x) \to f(y)$, the unique morphism between $f(x), f(y)$ in $\mathscr{E}Y$.

Inspecting the definition, we see $\mathscr{E} : \mathbf{Set} \to \mathbf{Cat}$ is a well-defined functor, called the *chaotic functor*.

Notice if a functor $F : \mathscr{A} \to \mathscr{B}$ has a chaotic target $\mathscr{B}$, then all the morphisms $f : X \to Y$ in $\mathscr{A}$ must be sent to the unique morphism $F(X) \to F(Y)$ in $\mathscr{B}$, hence F is dictated by the object map $F : \mathrm{Ob}(\mathscr{A}) \to \mathrm{Ob}(\mathscr{B})$. Such uniqueness specifies an adjunction $(\mathrm{Ob} \dashv \mathscr{E})$.

3.3 Pseudofunctor, Pseudotransform, and Modification

Definition 3.6 (Pseudofunctor)**.** Given two 2-categories $\mathscr{A}, \mathscr{B}$, a *pseudofunctor* $F : \mathscr{A} \to \mathscr{B}$ equips with an 0-cell map, functors between hom-categories $F_{X,Y} : \mathscr{A}(X,Y) \to \mathscr{B}(F(X), F(Y))$, an invertible 2-cell $\phi_{X,Y,Z}$ between compositions

$$\begin{tikzcd}
\mathscr{A}(Y,Z) \times \mathscr{A}(X,Y) \arrow[r, "F_{Y,Z} \times F_{X,Y}"] \arrow[d, "\circ"'] & \mathscr{B}(F(Y),F(Z)) \times \mathscr{B}(F(X),F(Y)) \arrow[d, "\circ"] \arrow[dl, Rightarrow, "\phi_{X,Y,Z}"'] \\
\mathscr{A}(X,Z) \arrow[r, "F_{X,Z}"'] & \mathscr{B}(F(X),F(Z))
\end{tikzcd}$$

and another invertible 2-cell ψ_X between identities

$$\begin{tikzcd}
F(X) \arrow[r, bend left, "\mathrm{id}_{F(X)}", ""{name=U, below}] \arrow[r, bend right, "F_{X,X}(\mathrm{id}_X)"', ""{name=D, above}] & F(X) \arrow[Rightarrow, from=U, to=D, "\psi_X"]
\end{tikzcd}$$

such that the following axioms hold:

(i) **(Associativity)** The following two pasting diagrams are equal:

$$\begin{tikzcd}
\mathscr{A}(Z,W) \times \mathscr{A}(Y,Z) \times \mathscr{A}(X,Y) \arrow[r, "\prod F_{-,-}"] \arrow[d, "\circ\times\mathrm{id}"'] & \mathscr{B}(F(Z),F(W)) \times \mathscr{B}(F(Y),F(Z)) \times \mathscr{B}(F(X),F(Y)) \arrow[d, "\circ\times\mathrm{id}"] \arrow[dl, Rightarrow, "\phi_{Y,Z,W}\times\mathrm{id}"'] \\
\mathscr{A}(Y,W) \times \mathscr{A}(X,Y) \arrow[r, "F_{Y,W} \times F_{X,Y}"] \arrow[d, "\circ"'] & \mathscr{B}(F(Y),F(W)) \times \mathscr{B}(F(X),F(Y)) \arrow[d, "\circ"] \arrow[dl, Rightarrow, "\phi_{X,Y,W}"'] \\
\mathscr{A}(X,W) \arrow[r, "F_{X,W}"'] & \mathscr{B}(F(X),F(W))
\end{tikzcd}$$

equals

$$\begin{tikzcd}
\mathscr{A}(Z,W) \times \mathscr{A}(Y,Z) \times \mathscr{A}(X,Y) \arrow[r, "\prod F_{-,-}"] \arrow[d, "\mathrm{id}\times\circ"'] & \mathscr{B}(F(Z),F(W)) \times \mathscr{B}(F(Y),F(Z)) \times \mathscr{B}(F(X),F(Y)) \arrow[d, "\mathrm{id}\times\circ"] \arrow[dl, Rightarrow, "\mathrm{id}\times\phi_{X,Y,Z}"'] \\
\mathscr{A}(Z,W) \times \mathscr{A}(X,Z) \arrow[r, "F_{Z,W} \times F_{X,Z}"] \arrow[d, "\circ"'] & \mathscr{B}(F(Z),F(W)) \times \mathscr{B}(F(X),F(Z)) \arrow[d, "\circ"] \arrow[dl, Rightarrow, "\phi_{X,Z,W}"'] \\
\mathscr{A}(X,W) \arrow[r, "F_{X,W}"'] & \mathscr{B}(F(X),F(W))
\end{tikzcd}$$

(ii) **(Unitality)** The following two unit diagrams commute:

$$\begin{tikzcd}
\mathscr{A}(X,Y) \times \mathscr{A}(X,X) \arrow[r, "F_{X,Y} \times F_{X,X}"] & \mathscr{B}(F(X),F(Y)) \times \mathscr{B}(F(X),F(X)) \arrow[dd, "\circ"] \arrow[ddl, Rightarrow, "\mathrm{id}*\psi_X"'] \\
\mathscr{A}(X,Y) \times * \arrow[u, "\mathrm{id}\times\mathrm{id}_X"] \arrow[d, "F_{X,Y}\times\mathrm{id}_{F(X)}"'] & \\
\mathscr{B}(F(X),F(Y)) \times \mathscr{B}(F(X),F(X)) \arrow[r, "\circ"'] & \mathscr{B}(F(X),F(Y))
\end{tikzcd}$$

$$
\begin{tikzcd}
\mathscr{A}(Y,Y)\times\mathscr{A}(X,Y) \arrow[r, "F_{Y,Y}\times F_{X,Y}"] & \mathscr{B}(F(Y),F(Y))\times\mathscr{B}(F(X),F(Y)) \arrow[dd, "\circ"] \arrow[ddl, Rightarrow, "\psi_Y * \mathrm{id}"'] \\
* \times \mathscr{A}(X,Y) \arrow[u, "\mathrm{id}_Y\times \mathrm{id}"] \arrow[d, "\mathrm{id}_{F(Y)}\times F_{X,Y}"'] & \\
\mathscr{B}(F(Y),F(Y))\times\mathscr{B}(F(X),F(Y)) \arrow[r, "\circ"'] & \mathscr{B}(F(X),F(Y))
\end{tikzcd}
$$

where $\mathrm{id}_X : * \to \mathscr{A}(X,X)$ is the functor indicating identity of X.

If ψ_X is identity 2-cell for all 0-cell X, then the pseudofunctor F is called *normal*.

Definition 3.7 (Pseudotransform)**.** Fix 2-categories $\mathscr{A},\mathscr{B}$, let $(F,\phi,\psi),(G,\varphi,\vartheta) : \mathscr{A}\to\mathscr{B}$ be two pseudofunctors. A *pseudotransform* $\mu : (F,\phi,\psi)\to(G,\varphi,\vartheta)$ collects 1-cells $\mu_X : F(X)\to G(X)$ for each 0-cell $X\in\mathscr{A}$, and for each 1-cell $f : X\to Y$ in $\mathscr{A}$, associates a 2-cell μf such that

$$
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, "Ff"'] & G(X) \arrow[d, "Gf"] \arrow[dl, Rightarrow, "\mu f"'] \\
F(Y) \arrow[r, "\mu_Y"'] & G(Y)
\end{tikzcd}
$$

commutes.

Furthermore, they satisfy the following axioms:

(i) (**Associativity**) Let $f : X\to Y$, $g : Y\to Z$ be 1-cells in $\mathscr{A}$, the following two pasting diagrams are equal:

$$
\begin{tikzcd}
F(X) \arrow[rr, "\mu_X"] \arrow[dr, "Ff"] \arrow[dd, "F(g\circ f)"'] & & G(X) \arrow[dl, Rightarrow, "\mu f"'] \arrow[dr, "Gf"] & \\
\arrow[r, phantom, "\overset{\phi_{g,f}}{\Longleftarrow}"] & F(Y) \arrow[rr, "\mu_Y"] \arrow[dl, "Fg"] & & G(Y) \arrow[dlll, Rightarrow, "\mu g"'] \arrow[dl, "Gg"] \\
F(Z) \arrow[rr, "\mu_Z"'] & & G(Z) &
\end{tikzcd}
=
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[dd, "F(g\circ f)"'] & G(X) \arrow[ddl, Rightarrow, "\mu(g\circ f)"'] \arrow[dd, "G(g\circ f)"'] \arrow[dr, "Gf"] & \\
 & \arrow[r, phantom, "\overset{\varphi_{g,f}}{\Longleftarrow}"] & G(Y) \arrow[dl, "Gg"] \\
F(Z) \arrow[r, "\mu_Z"'] & G(Z) &
\end{tikzcd}
$$

(ii) (**Unitality**) The following pasting diagrams are equal:

$$
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, bend right=50, "F(\mathrm{id}_X)"'] \arrow[d, bend left=50, "\mathrm{id}_{F(X)}"] \arrow[d, phantom, "\Downarrow\psi_X"] & G(X) \arrow[d, "\mathrm{id}_{G(X)}"] \\
F(X) \arrow[r, "\mu_X"'] & G(X)
\end{tikzcd}
=
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, "F(\mathrm{id}_X)"'] & G(X) \arrow[dl, Rightarrow, "\mu(\mathrm{id}_X)"'] \arrow[d, bend right=50, "G(\mathrm{id}_X)"'] \arrow[d, bend left=50, "\mathrm{id}_{G(X)}"] \arrow[d, phantom, "\Downarrow\vartheta_X"] \\
F(X) \arrow[r, "\mu_X"'] & G(X)
\end{tikzcd}
$$

Definition 3.8 (Modification)**.** Continue with the same notation as above. Let $\mu,\nu : (F,\phi,\psi)\to(G,\varphi,\vartheta)$ be two pseudotransforms, a *modification* $\lambda : \mu\Rightarrow\nu$ is a collection of 2-cells $\lambda_X : \mu_X\Rightarrow\nu_X$ for each 0-cell

$X \in \mathscr{A}$, such that the following two pasting diagrams are equal, for each 1-cell $f : X \to Y$ in $\mathscr{A}$:

$$
\begin{tikzcd}
F(X) \arrow[r, bend left, "\mu_X"] \arrow[r, bend right, "\nu_X"'] \arrow[r, phantom, "\Downarrow \lambda_X"] \arrow[d, "Ff"'] & G(X) \arrow[d, "Gf"] \arrow[dl, Rightarrow, "\nu f"] \\
F(Y) \arrow[r, "\nu_Y"'] & G(Y)
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
F(X) \arrow[r, "\mu_X"] \arrow[d, "Ff"'] & G(X) \arrow[d, "Gf"] \arrow[dl, Rightarrow, "\mu f"'] \\
F(Y) \arrow[r, bend left, "\mu_Y"] \arrow[r, bend right, "\nu_Y"'] \arrow[r, phantom, "\Downarrow \lambda_Y"] & G(Y)
\end{tikzcd}
$$

3.4 G-Categories

Definition 3.9. A small category $\mathscr{A}$ is a *G-category*, if the object and morphism sets $\mathrm{Ob}(\mathscr{A}), \mathrm{Mor}(\mathscr{A})$ are both (left) G-sets, such that the following four structure maps (S, T, I, C) (stands for *source, target, identity, and composition*) are all G-equivariant:

- $S, T : \mathrm{Mor}(\mathscr{A}) \to \mathrm{Ob}(\mathscr{A})$ by $S(X \xrightarrow{f} Y) := X$, and $T(X \xrightarrow{f} Y) := Y$.
- $I : \mathrm{Ob}(\mathscr{A}) \to \mathrm{Mor}(\mathscr{A})$ by $I(X) := \mathrm{id}_X$.
- $C : \mathrm{Mor}(\mathscr{A}) \times_{S,T} \mathrm{Mor}(\mathscr{A}) \to \mathrm{Mor}(\mathscr{A})$ by $C(g, f) := g \circ f$.

Here, $\mathrm{Mor}(\mathscr{A}) \times_{S,T} \mathrm{Mor}(\mathscr{A}) := \{(g, f) \in \mathrm{Mor}(\mathscr{A})^2 \mid S(g) = T(f)\}$, precisely the pair of morphisms that're composable, and it endows G-set structure via restricting the diagonal G-action on $\mathrm{Mor}(\mathscr{A})^2$. This is well-defined due to the assumption S, T are G-maps.

Another equivalent characterization can be viewed as a group homomorphism $G \to \mathrm{Aut}_{\mathbf{Cat}}(\mathscr{A})$, which assigns each $g \in G$ an automorphism functor on $\mathscr{A}$ such that composition respects G's multiplication structure, and identity gets sent to the identity functor. So, from now on we view each element $g \in G$ as a functor, whenever a category is specified as a G-category.

Definition 3.10. Let $\mathbf{Cat}_G$ denotes the 2-category of small G-categories, with (nonequivariant) functors as 1-cells, natural transformations as 2-cells. The hom-category between any two G-categories $\mathbf{Cat}_G(\mathscr{A}, \mathscr{B})$ also has a natural G-action via conjugation. Which, the G-fixed subcategory of the hom-category $[\mathbf{Cat}_G(\mathscr{A}, \mathscr{B})]^G$ precisely describes all G-functor, or the functors $F : \mathscr{A} \to \mathscr{B}$ such that $g \circ F = F \circ g$ for all $g \in G$.

Another important notion is the twisted G-action on products when given a group homomorphism $\alpha : G \to \Sigma_n$. Given a category $\mathscr{A}$, the n-fold product $\mathscr{A}^{\mathbf{n}}$ has a natural left Σ_n-action via permuting the coordinates; if $\mathscr{A}$ is a G-category, $\mathscr{A}^{\mathbf{n}}$ also endows a natural diagonal G-action. The two coordinates to give a natural left $(G \times \Sigma_n)$-action on $\mathscr{A}^{\mathbf{n}}$.

Definition 3.11. Let $\mathscr{A}^{\mathbf{n}^\alpha}$ be a G-category with the underlying category being $\mathscr{A}^{\mathbf{n}}$, with the G-action defined as follow:

$$g \cdot_\alpha (X_1, ..., X_n) := (g \cdot X_{\alpha(g)^{-1}(1)}, ..., g \cdot X_{\alpha(g)^{-1}(n)})$$

Another characterization can be understood as follow: Given the graph subgroup $\Gamma_\alpha := \{(g, \alpha(g)) \in G \times \Sigma_n\}_{g \in G}$, the projection $\pi_G : G \times \Sigma_n \twoheadrightarrow G$ restricts to an isomorphism $\Gamma_\alpha \xrightarrow{\sim} G$. The α-twisted action is induced by the pullback along this isomorphism.

4 Operads, Operadic Algebra, and Operad $\mathscr{P}$

4.1 Symmetric Monoidal Categories

Definition 4.1. A category $\mathcal{V}$ is a *symmetric monoidal category*, if there is a bifunctor $\otimes : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$ (called *monoidal product*), and an object $\kappa \in \mathcal{V}$ (called *unit/monoidal unit*), together with the following natural isomorphisms:

- The *associator* $a : (\_ \otimes \_) \otimes \_ \xrightarrow{\sim} \_ \otimes (\_ \otimes \_)$ of functors $\mathcal{V}^3 \xrightarrow{\sim} \mathcal{V}$.
- The *left unit* $\lambda : \kappa \otimes \_ \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ and the *right unit* $\rho : \_ \otimes \kappa \xrightarrow{\sim} \mathrm{Id}_{\mathcal{V}}$ of functors $\mathscr{V} \to \mathscr{V}$.
- The *Symmetric Braiding* $B : \_{1} \otimes \_{2} \xrightarrow{\sim} \_{2} \otimes \_{1}$, with extra property $B_{Y,X} : Y \otimes X \to X \otimes Y$ equals to $B^{-1}_{X,Y} : X \otimes Y \to Y \otimes X$.

They also satisfy the following axioms:

- *Triangle Axiom:*

$$\begin{tikzcd}
(X \otimes \kappa) \otimes Y \arrow[rr, "a_{X,\kappa,Y}"] \arrow[dr, "\rho_X \otimes \mathrm{id}_Y"'] & & X \otimes (\kappa \otimes Y) \arrow[dl, "\mathrm{id}_X \otimes \lambda_Y"] \\
& X \otimes Y &
\end{tikzcd}$$

- *Pentagon Axiom:*

$$\begin{tikzcd}
((X \otimes Y) \otimes Z) \otimes W \arrow[rr, "a_{X\otimes Y,Z,W}"] \arrow[d, "a_{X,Y,Z} \otimes \mathrm{id}_W"'] & & (X \otimes Y) \otimes (Z \otimes W) \arrow[d, "a_{X,Y,Z\otimes W}"] \\
(X \otimes (Y \otimes Z)) \otimes W \arrow[dr, "a_{X,Y\otimes Z,W}"'] & & X \otimes (Y \otimes (Z \otimes W)) \\
& X \otimes ((Y \otimes Z) \otimes W) \arrow[ur, "\mathrm{id}_X \otimes a_{Y,Z,W}"'] &
\end{tikzcd}$$

- *Hexagon Axiom:*

$$\begin{tikzcd}
(X \otimes Y) \otimes Z \arrow[r, "a_{X,Y,Z}"] \arrow[d, "B_{X,Y} \otimes \mathrm{Id}_Z"'] & X \otimes (Y \otimes Z) \arrow[r, "B_{X,Y\otimes Z}"] & (Y \otimes Z) \otimes X \arrow[d, "a_{Y,Z,X}"] \\
(Y \otimes X) \otimes Z \arrow[r, "a_{Y,X,Z}"'] & Y \otimes (X \otimes Z) \arrow[r, "\mathrm{id}_Y \otimes B_{X,Z}"'] & Y \otimes (Z \otimes X)
\end{tikzcd}$$

Based on *Maclane's Coherence Theorem*, the above two axioms guarantees all finite sequences of a, λ, ρ and their inverses agree, namely there is a unique isomorphism $P_1 \to P_2$, where P_i are (possibly) different orders of performing product on $X_1, ..., X_n \in \mathcal{V}$. For details, cf. [11].

4.2 Operads, Operadic Algebras

Definition 4.2. Given symmetric monoidal category $\mathcal{V}$, a Σ *operad* $\mathcal{C}$ in $\mathcal{V}$ is a collection of objects $\{\mathcal{C}(j) \in \mathcal{V}\}_{j\in\mathbb{N}}$, each $\mathcal{C}(j)$ with a right Σ_j-action (a group homomorphism $\Sigma_j^{\mathrm{op}} \to \mathrm{Aut}_{\mathcal{V}}(\mathcal{C}(j))$), a *unit map* $\eta : 1 \to \mathcal{C}(1)$, together with composition maps

$$\gamma : \mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes ... \otimes \mathcal{C}(j_k) \to \mathcal{C}(j_1 + ... + j_k)$$

that satisfies the following axioms:

- *Unit:*

$$\begin{tikzcd}
\mathcal{C}(j)\otimes\kappa^{\otimes j} \arrow[rr, "\sim"] \arrow[dr, "\mathrm{id}_{\mathcal{C}(j)}\otimes\eta^{\otimes j}"'] & & \mathcal{C}(j)\ , \\
& \mathcal{C}(j)\otimes\mathcal{C}(1)^{\otimes j} \arrow[ur, "\gamma"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\kappa\otimes\mathcal{C}(j) \arrow[rr, "\sim"] \arrow[dr, "\eta\otimes\mathrm{id}_{\mathcal{C}(j)}"'] & & \mathcal{C}(j) \\
& \mathcal{C}(1)\otimes\mathcal{C}(j) \arrow[ur, "\gamma"'] &
\end{tikzcd}$$

- *Associativity:* Fix natural numbers k, $j=\sum_{s=1}^{k} j_s$, define index $g_s := j_1+...+j_s$ for $1\le s\le k$. For each $i_1,...,i_{g_k}=i_j$, with sum $i=\sum_{r=1}^{j} i_r$, and $h_s := i_{g_{(s-1)}+1}+...+i_{g_s}$ (so $i=\sum_{s=1}^{k} h_s$ also), the following diagram commutes:

$$\begin{tikzcd}
\left[\mathcal{C}(k)\otimes\left(\bigotimes_{s=1}^{k}\mathcal{C}(j_s)\right)\right]\otimes\left(\bigotimes_{r=1}^{j}\mathcal{C}(i_r)\right) \arrow[r, "\gamma\otimes\mathrm{id}"] \arrow[dd, "\text{shuffle}"'] & \mathcal{C}(j)\otimes\left(\bigotimes_{r=1}^{j}\mathcal{C}(i_r)\right) \arrow[d, "\gamma"] \\
& \mathcal{C}(i) \\
\mathcal{C}(k)\otimes\left[\bigotimes_{s=1}^{k}\left(\mathcal{C}(j_s)\otimes\left(\bigotimes_{q=1}^{j_s}\mathcal{C}(i_{g_{(s-1)}+q})\right)\right)\right] \arrow[r, "\mathrm{id}_{\mathcal{C}(k)}\otimes(\bigotimes_s\gamma)"'] & \mathcal{C}(k)\otimes\left(\bigotimes_{s=1}^{k}\mathcal{C}(h_s)\right) \arrow[u, "\gamma"']
\end{tikzcd}$$

- *Equivariance:*

  - Given $\sigma\in\Sigma_k$, $j_1+...+j_k=\Sigma_j$, let $\sigma(j_1,...,j_k)\in\Sigma_j$ denotes "permutation of k-blocks of letters", by $(j_1,...,j_k)\mapsto(j_{\sigma^{-1}(1)},...,j_{\sigma^{-1}(k)})$.

    Example 4.3. Given $j=5=2+3$, the permutation $(1\ 2)\in\Sigma_2$, $\sigma(2,3)$ is the permutation $\langle 1,2;3,4,5\rangle\mapsto\langle 3,4,5;1,2\rangle$ (namely changing the blocks' order, without changing the order in each block).

    Then, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C}(k)\otimes\left(\bigotimes_{r=1}^{k}\mathcal{C}(j_r)\right) \arrow[r, "\sigma\otimes\ \text{shuffle}_{\sigma^{-1}}"] \arrow[d, "\gamma"'] & \mathcal{C}(k)\otimes\left(\bigotimes_{r=1}^{k}\mathcal{C}(j_{\sigma(r)})\right) \arrow[d, "\gamma"] \\
\mathcal{C}(j) \arrow[r, "\sigma(j_{\sigma(1)},...,j_{\sigma(k)})"'] & \mathcal{C}(j)
\end{tikzcd}$$

  - Given $\tau_r\in\Sigma_{j_r}$, denote $\tau_1\oplus...\oplus\tau_k\in\Sigma_j$ by letting each τ_r permuting the j_rth block in j (for instance, let τ_1 acts on $1,...,j_1$; let τ_2 acts on $j_1+1,...,j_1+j_2$, etc.). Then, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C}(k)\otimes\left(\bigotimes_{r=1}^{k}\mathcal{C}(j_r)\right) \arrow[r, "\mathrm{id}_{\mathcal{C}(k)}\otimes(\bigotimes_r\tau_r)"] \arrow[d, "\gamma"'] & \mathcal{C}(k)\otimes\left(\bigotimes_{r=1}^{k}\mathcal{C}(j_r)\right) \arrow[d, "\gamma"] \\
\mathcal{C}(j) \arrow[r, "\tau_1\oplus...\oplus\tau_k"'] & \mathcal{C}(j)
\end{tikzcd}$$

If $\mathcal{C}(0)=\kappa$ (with $\gamma:\mathcal{C}(0)\to\mathcal{C}$ being $\mathrm{id}_{\mathcal{C}(0)}$), the operad is *reduced.*

Definition 4.4. Let $\mathcal{C},\mathcal{C}'$ be two operads on symmetric monoidal category $\mathcal{V}$, a *morphism of operads* $\psi:\mathcal{C}\to\mathcal{C}'$, is a sequence of Σ_j-equivariant morphisms $\psi_j:\mathcal{C}(j)\to\mathcal{C}'(j)$ in $\mathcal{V}$ (i.e., $\sigma^{-1}\circ\psi_j\circ\sigma=\psi_j$, for all

$\sigma \in \Sigma_j$), such that it preserves the operad's structure morphism:

$$\begin{tikzcd}
\mathcal{C}(k) \otimes \mathcal{C}(j_1) \otimes ... \otimes \mathcal{C}(j_k) \arrow[r, "\gamma"] \arrow[d, "\psi_k \otimes \psi_{j_1} \otimes ... \otimes \psi_{j_k}"'] & \mathcal{C}(j_1 + ... + j_k) \arrow[d, "\psi_{j_1+...+j_k}"] \\
\mathcal{C}'(k) \otimes \mathcal{C}'(j_1) \otimes .. \otimes \mathcal{C}'(j_k) \arrow[r, "\gamma'"'] & \mathcal{C}'(j_1 + ... + j_k)
\end{tikzcd}$$

Definition 4.5. Given an operad $\mathcal{C}$ over a symmetric monoidal category $\mathcal{V}$, a $\mathcal{C}$-*algebra* is an object $A \in \mathcal{V}$, with collections of morphisms $\theta : \mathcal{C}(j) \otimes A^j \to A$, that satisfies the following axioms:

- *Associativity:* Given $j = \sum_{i=1}^{k} j_i$, the following diagram commutes:

$$\begin{tikzcd}
\Big[C(k) \otimes \Big(\bigotimes_{i=1}^{k} C(j_i)\Big)\Big] \otimes A^{\otimes j} \arrow[r, "\gamma \otimes \mathrm{id}_{A^{\otimes j}}"] \arrow[dd, "\text{shuffle}"'] & C(j) \otimes A^{\otimes j} \arrow[dr, "\theta"] & \\
& & A \\
C(k) \otimes \Big[\bigotimes_{i=1}^{k} \big(C(j_i) \otimes A^{\otimes j_i}\big)\Big] \arrow[r, "\mathrm{id}_{C(k)} \otimes \theta^k"'] & C(k) \otimes A^{\otimes k} \arrow[ur, "\theta"'] &
\end{tikzcd}$$

(Note: This part uses the isomorphism $A^{\otimes j} \cong \bigotimes_{i=1}^{k} A^{\otimes j_i}$).

- *Unit:* The following diagram commutes:

$$\begin{tikzcd}
\kappa \otimes A \arrow[rr, "\sim"] \arrow[dr, "\eta \otimes \mathrm{id}_A"'] & & A \\
& C(1) \otimes A \arrow[ur, "\theta"'] &
\end{tikzcd}$$

- *Eauivariance:* Given any permutation $\sigma \in \Sigma_j$, the following diagram commutes:

$$\begin{tikzcd}
C(j) \otimes A^{\otimes j} \arrow[rr, "\sigma \otimes \sigma^{-1}"] \arrow[dr, "\theta"'] & & C(j) \otimes A^{\otimes j} \arrow[dl, "\theta"] \\
& A &
\end{tikzcd}$$

4.3 Permutativity Operad $\mathscr{P}$

We first starts by introducing a relevant operad in **Set**: The *associativity operad* $\mathscr{M}$ in **Set** is defined as $\mathscr{M}(j) := \Sigma_j$ (the symmetric group for cardinality j), which endows a natural right Σ_j-action via right multiplication. The operadic structure map is given by

$$\gamma : \Sigma_k \times \prod_{r=1}^{k} \Sigma_{j_r} \to \Sigma_j, \quad \gamma(\sigma; \tau_1, ..., \tau_k) := \tau_{\sigma^{-1}(1)} \oplus ... \oplus \tau_{\sigma^{-1}(k)} \circ \sigma(j_{\sigma(1)}, ..., j_{\sigma(k)}) \tag{4.6}$$

where $j = \sum_r j_r$, $\sigma(j_1, ..., j_k) \in \Sigma_j$ denotes the block permutation of $j_1, ..., j_k$ within j (in order), and $\tau_1 \oplus ... \tau_k$ is the image of the embedding $\oplus : \prod_{r=1}^{k} \Sigma_{j_r} \to \Sigma_j$, by viewing the rth block as finite set of cardinality j_r.

Inspecting the definitions, the algebra over $\mathscr{M}$ precisely describes the set theoretic monoids. More details can be found in **(Yet to be determined...)**

Now, recall that the chaotic functor $\mathscr{E} : \mathbf{Set} \to \mathbf{Cat}$ is a right adjoint of $\mathrm{Ob} : \mathbf{Cat} \to \mathbf{Set}$, in particular it preserves products. Hence, the chaotic categorification $\mathscr{P}(j) := \mathscr{E}\mathscr{M}(j)$ also has a natural operadic structure, via the identification

$$\mathscr{E}\mathscr{M}(k) \times \prod_{r=1}^{k} \mathscr{E}\mathscr{M}(j_r) \cong \mathscr{E}\left(\mathscr{M}(k) \times \prod_{r=1}^{k} \mathscr{M}(j_r)\right) \xrightarrow{\mathscr{E}\gamma} \mathscr{E}\mathscr{M}(j). \tag{4.7}$$

This is called the *permutativity operad in* **Cat**, or the *Barratt-Eccles operad* as credited to the authors who proposed it. As an abuse of notation, the operadic structure map of $\mathscr{P}$ is also denoted γ.

It is known in **(Yet to be determined...check sym mon cat)** that the algebras over $\mathscr{P}$ is isomorphic to the permutative categories. Later on, in **(Yet to be determined...)** the concept of pseudoalgebra over $\mathscr{P}$ is proposed, and in [9] it specified an equivalence between the $\mathscr{P}$-pseudoalgebras and the symmetric monoidal categories, by relaxing the strict associativity/unity in $\mathscr{P}$-algebras to be only up to coherent natural isomorphisms.

5 G-Operad $\mathscr{P}_G$ and its Pseudoalgebra

5.1 The Barratt-Eccels G-Operad $\mathscr{P}_G$

If view each $\mathscr{P}(j)$ as a G-trivial G-category, it's also an operad in $\mathbf{Cat}_G$. Its underlying algebra is called the *naive permutative G-category*, as specified in **(Yet to be determined...)** it produces a naive G-spectrum under equivariant infinite loop space theory. In [3], a richer equivariant structure is introduced which produces the *equivariant Barratt Eccles G-operad $\mathscr{P}_G$* as follow:

Definition 5.1. The jth category $\mathscr{P}_G(j) := \mathbf{Cat}(\mathscr{E}G, \mathscr{P}(j))$, with the operadic structure map determined by the fact that $\mathbf{Cat}(\mathscr{E}G, \_)$ preserves product:

$$\mathbf{Cat}(\mathscr{E}G, \mathscr{P}(k)) \times \prod_{r=1}^{k} \mathbf{Cat}(\mathscr{E}G, \mathscr{P}(j_r)) \cong \mathbf{Cat}\left(\mathscr{E}G, \mathscr{P}(k) \times \prod_{r=1}^{k} \mathscr{P}(j_r)\right) \xrightarrow{\gamma\circ\_} \mathbf{Cat}(\mathscr{E}, \mathscr{P}(j)) \tag{5.2}$$

Abuse of notation, we'll denote the operadic structure map as γ again.

Another formulation can be done by defining $P_G(j) := \mathbf{Set}(G, \Sigma_j)$ in $\mathbf{Set}$, then utilize chaotic functor promoting this to a chaotic operad $\mathscr{E}\mathbf{Set}(G, \Sigma) \cong \mathscr{P}_G$.

5.2 The 2-Category $\mathscr{P}_G$-PsAlg (Need to check what's the equivariance for natural transform)

We mostly follow [9] in defining $\mathscr{P}_G$-pseudoalgebras.

Definition 5.3 ($\mathscr{P}_G$-pseudoalgebra)**.** A $\mathscr{P}_G$-pseudoalgebra is a G-category $\mathscr{A}$ and operad action functors

$$\theta_j : \mathscr{P}_G(j) \times \mathscr{A}^j \to \mathscr{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$\begin{tikzcd} \mathscr{A} \arrow[r, "\iota"] \arrow[dr, "\mathrm{Id}_{\mathscr{A}}"'] & \mathscr{P}_G(1) \times \mathscr{A} \arrow[d, "\theta_1"] \arrow[dl, phantom, "\Leftarrow \phi_1"] \\ & C \end{tikzcd} \tag{5.4}$$

An invertible natural transformation $\phi = \phi(k; j_1, ..., j_k)$ for each diagram

$$\begin{tikzcd} \mathscr{P}_G(k) \times \left(\prod_{r=1}^{k} \mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[r, "\mathrm{id}\times\prod_r \theta_{j_r}"] \arrow[dd, "\text{shuffle}"'] & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[d, "\theta_k"] \arrow[ddl, Rightarrow, "\phi"] \\ & \mathscr{A} \\ \mathscr{P}_G(k) \times \prod_{r=1}^{k} \mathscr{P}_G(j_r) \times \mathscr{A}^{j_+} \arrow[r, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(j_+) \times \mathscr{A}^{j_+} \arrow[u, "\theta_{j_+}"'] \end{tikzcd} \tag{5.5}$$

such that the following strict equivariance hold for θ_j and ϕ:

(i) **(Equivariance of θ_j)** The following diagram commutes for $(g, \sigma) \in G \times \Sigma_j$:

$$\begin{tikzcd} \mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, "(g,\sigma)\times(g,\sigma)^{-1}"] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{A}^j \arrow[d, "\theta_j"] \\ \mathscr{A} \arrow[r, "g^{-1}"'] & \mathscr{A} \end{tikzcd} \tag{5.6}$$

I think if assuming the commutation for G also, the case $j = 1$ implies θ_j collapses all G-action on $\mathscr{A}$ to be trivial

(ii) **(Equivariance of ϕ)** The following equalities hold when given $g \in G$, $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, ..., j_k) = \phi(k; j_{\sigma(1)}, ..., j_{\sigma_k}) \circ ((g, \sigma) \times (g, \sigma)^{-1})$$

$$\phi(k; j_1, ..., j_k) = \phi(k; j_1, ..., j_k) \circ (\mathrm{id} \times \prod_r ((g, \tau_r) \times (g, \tau_r)^{-1}))$$

They also need to satisfy the following coherence properties:

(iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \leq r \leq k$ and $1 \leq s \leq j_r$:

$$\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \prod_s \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right)\right) \arrow[r, "\mathrm{id}\times\prod(\mathrm{id}\times\theta_{i_{r,s}})"] \arrow[dd, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] \arrow[dd, "\gamma"'] & \\
 & & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[dl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"] \\
\mathscr{P}_G(j_+) \times \prod_{r,s} \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right) \arrow[r, "\mathrm{id}\times\prod\theta_{i_{r,s}}"] \arrow[dr, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(j_+) \times \mathscr{A}^{j_+} \arrow[d, Rightarrow, "\phi"] \arrow[dr, "\theta_{j_+}"] & \\
 & \mathscr{P}_G(i_+) \times \mathscr{A}^{i_+} \arrow[r, "\theta_{j_+}"'] & \mathscr{A}
\end{tikzcd}$$

equals

$$\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \prod_s \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right)\right) \arrow[r, "\mathrm{id}\times\prod(\mathrm{id}\times\prod\theta_{i_{r,s}})"] \arrow[dr, "\mathrm{id}\times\prod(\gamma\times\mathrm{id})"'] \arrow[dd, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[d, Rightarrow, "\mathrm{id}\times\prod\phi"] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & \\
 & \mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(i_r) \times \mathscr{A}^{i_r}\right) \arrow[r, "\mathrm{id}\times\prod\gamma"'] \arrow[dd, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[ddl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"] \\
\mathscr{P}_G(j_+) \times \prod_{r,s} \left(\mathscr{P}_G(i_{r,s}) \times \mathscr{A}^{i_{r,s}}\right) \arrow[dr, "\gamma\times\mathrm{id}"'] & & \\
 & \mathscr{P}_G(i_+) \times \mathscr{A}^{i_+} \arrow[r, "\theta_{i_+}"'] & \mathscr{A}
\end{tikzcd}$$

(commuting squares are operadic associativity)

(iv) **(Operadic Identity)** The following two pairs of two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{P}_G(k) \times \mathscr{A}^k \arrow[r, "\mathrm{id}\times\iota^k"] & \mathscr{P}_G(k) \times (\mathscr{P}_G(1) \times \mathscr{A})^k \arrow[r, "\mathrm{id}\times\theta_1^k"] \arrow[d, "\mathrm{shuffle}"'] & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[dd, "\theta_k"] \arrow[ddl, phantom, "\Swarrow \phi"] \\
& \mathscr{P}_G(k) \times \mathscr{P}_G(1)^k \times \mathscr{A}^k \arrow[d, "\gamma\times\mathrm{id}"'] & \\
& \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[r, "\theta_k"'] & \mathscr{A}
\end{tikzcd}
$$

equals

$$
\begin{tikzcd}
& & \mathscr{P}_G(k) \times (\mathscr{P}_G(1) \times \mathscr{A})^k \arrow[dr, "\mathrm{id}\times\theta_1^k"] \arrow[d, phantom, "\Downarrow \mathrm{id}\times\prod\phi"] & & \\
& \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[ur, "\mathrm{id}\times\iota^k"] \arrow[rr, "\mathrm{id}"'] \arrow[d, "\mathrm{id}\times\iota^k"'] \arrow[dr, "\iota^k"] & {} & \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[r, "\theta_k"] & \mathscr{A} \\
& \mathscr{P}_G(k) \times (\mathscr{P}_G(1) \times \mathscr{A})^k \arrow[r, "\mathrm{shuffle}"'] & \mathscr{P}_G(k) \times \mathscr{P}_G(1)^k \times \mathscr{A}^k \arrow[ur, "\gamma\times\mathrm{id}"'] & &
\end{tikzcd}
$$

(uses operadic associativity)

Also:

$$
\begin{tikzcd}
\mathscr{P}_G(k) \times \mathscr{A}^k \arrow[r, "\iota"] \arrow[dr, "\mathrm{id}"'] & \mathscr{P}_G(1) \times \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[r, "\mathrm{id}\times\theta_k"] \arrow[d, "\gamma\times\mathrm{id}"'] & \mathscr{P}_G(1) \times \mathscr{A} \arrow[d, "\theta_1"] \arrow[dl, phantom, "\Swarrow \phi"] \\
& \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[r, "\theta_k"'] & \mathscr{A}
\end{tikzcd}
$$

equals

$$
\begin{tikzcd}
\mathscr{P}_G(k) \times \mathscr{A}^k \arrow[r, "\iota"] \arrow[d, "\theta_k"'] & \mathscr{P}_G(1) \times \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[d, "\mathrm{id}\times\theta_k"] \\
\mathscr{A} \arrow[r, "\iota"] \arrow[dr, "\mathrm{id}"'] & \mathscr{P}_G(1) \times \mathscr{A} \arrow[d, "\theta_1"] \arrow[l, phantom, shift left=3, "\Leftarrow \phi_1"'] \\
& \mathscr{A}
\end{tikzcd}
$$

(Uses operadic unitality)

Definition 5.7 ($\mathscr{P}_G$-pseudomorphism)**.** A $\mathscr{P}_G$-pseudomorphism $(\mathbb{F}, \zeta) : (\mathscr{A}, \theta, \phi) \to (\mathscr{B}, \vartheta, \varphi)$ consists of functor $\mathbb{F} : \mathscr{A} \to \mathscr{B}$, such that for each j in the follogin diagram commutes up to an invertible natural transformation ζ_j:

$$
\begin{tikzcd}
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, "\mathrm{id}\times\mathbb{F}^j"] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, phantom, "\Swarrow \zeta_j"] \\
\mathscr{A} \arrow[r, "\mathbb{F}"'] & \mathscr{B}
\end{tikzcd}
$$

Moreover, they satisfy the following properties:

(i) **(Equivariance of ζ)** For each j,$\zeta_j = \zeta_j \circ (\sigma \times \sigma^{-1})$ for all $\sigma \in \Sigma_j$. **Check G-equivariance**

(ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{A} \arrow[r, "\mathbb{F}"] \arrow[d, "\iota"'] & \mathscr{B} \arrow[d, "\iota"'] \arrow[ddr, bend left, "\mathrm{id}"] & \\
\mathscr{P}_G(1) \times \mathscr{A} \arrow[r, "\mathrm{id}\times\mathbb{F}"] \arrow[dr, "\theta_1"'] & \mathscr{P}_G(1) \times \mathscr{B} \arrow[dr, "\vartheta_1"'] \arrow[r, phantom, "\overset{\varphi_1}{\Rightarrow}"] & \\
& \mathscr{A} \arrow[r, "\mathbb{F}"'] \arrow[u, Rightarrow, "\zeta_1^{-1}"'] & \mathscr{B}
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\mathscr{A} \arrow[r, "\mathbb{F}"] \arrow[d, "\iota"'] \arrow[dr, "\mathrm{id}"] \arrow[d, phantom, shift left=4, "\overset{\phi_1}{\Rightarrow}"] & \mathscr{B} \arrow[dr, "\mathrm{id}"] & \\
\mathscr{P}_G(1) \times \mathscr{A} \arrow[r, "\theta_1"'] & \mathscr{A} \arrow[r, "\mathbb{F}"'] & \mathscr{B}
\end{tikzcd}
$$

(iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rr, "\mathrm{id}\times\prod(\mathrm{id}\times\mathbb{F}^{j_r})"] \arrow[dd, "\gamma"'] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & & \mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{B}^{j_r}\right) \arrow[dl, Rightarrow, "\mathrm{id}\times\prod\zeta_{j_r}"] \arrow[dr, "\mathrm{id}\times\prod\vartheta_{j_r}"] & \\
& \mathscr{P}_G(k) \times \mathscr{A}^k \arrow[rr, "\mathrm{id}\times\mathbb{F}^k"'] \arrow[dl, Rightarrow, "\phi"] \arrow[dd, "\theta_k"'] & & \mathscr{P}_G(k) \times \mathscr{B}^k \arrow[ddll, Rightarrow, "\zeta_k"] \arrow[dd, "\vartheta_k"] \\
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[dr, "\theta_j"'] & & & \\
& \mathscr{A} \arrow[rr, "\mathbb{F}"'] & & \mathscr{B}
\end{tikzcd}
$$

equals

$$
\begin{tikzcd}
\mathscr{P}_G(k) \times \prod_r \left(\mathscr{P}_G(j_r) \times \mathscr{A}^{j_r}\right) \arrow[rr, "\mathrm{id}\times\prod(\mathrm{id}\times\mathbb{F}^{j_r})"] \arrow[dd, "\gamma"'] & & \mathscr{P}_G(k) \times \prod \left(\mathscr{P}_G(j_r) \times \mathscr{B}^{j_r}\right) \arrow[dd, "\gamma"'] \arrow[dr, "\mathrm{id}\times\prod\theta_{j_r}"] & \\
& & & \mathscr{P}_G(k) \times \mathscr{B}^k \arrow[dl, Rightarrow, "\varphi"] \arrow[dd, "\vartheta_k"] \\
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[rr, "\mathrm{id}\times\mathbb{F}^j"] \arrow[dr, "\theta_j"'] & & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[dl, Rightarrow, "\zeta_j"] \arrow[dr, "\vartheta_j"] & \\
& \mathscr{A} \arrow[rr, "\mathbb{F}"'] & & \mathscr{B}
\end{tikzcd}
$$

Definition 5.8 ($\mathscr{P}_G$-2-cells)**.** Let $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : (\mathscr{A}, \theta, \phi) \to (\mathscr{B}, \vartheta, \varphi)$ be two $\mathscr{P}_G$-pseudomorphisms, a $\mathscr{P}_G$-2-cell $\lambda : (\mathbb{F}, \zeta) \Rightarrow (\mathbb{G}, \xi)$ is a natural transformation $\lambda : \mathbb{F} \to \mathbb{G}$, such that the following two pasting diagrams are equal:

$$
\begin{tikzcd}
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, bend left=50, "\mathrm{id}\times\mathbb{F}^j"] \arrow[r, bend right=50, "\mathrm{id}\times\mathbb{G}^j"'] \arrow[r, Rightarrow, "\mathrm{id}\times\lambda^j"] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, Rightarrow, "\xi_j"] \\
\mathscr{A} \arrow[r, "\mathbb{G}"'] & \mathscr{B}
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\mathscr{P}_G(j) \times \mathscr{A}^j \arrow[r, "\mathrm{id}\times\mathbb{F}^j"] \arrow[d, "\theta_j"'] & \mathscr{P}_G(j) \times \mathscr{B}^j \arrow[d, "\vartheta_j"] \arrow[dl, Rightarrow, "\zeta_j"'] \\
\mathscr{A} \arrow[r, bend left=40, "\mathbb{F}"] \arrow[r, bend right=40, "\mathbb{G}"'] \arrow[r, Rightarrow, "\lambda"'] & \mathscr{B}
\end{tikzcd}
$$

6 Category $\mathscr{F}_G$ and its Pseudoalgebras

Definition 6.1. $\mathscr{F}_G$ denotes the *category of based finite sets*, where the objects are $\mathbf{n}^\alpha$, set wise being the finite set $\{0, 1, ..., n\}$ with 0 fixed as the basepoint, and $\alpha : G \to \Sigma_n$ is a based G-action on the set.

The homset $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ consists of all the based set map, which the homset also endows with a G-action via conjugation. From now on, we'll view $\mathscr{F}_G$ as a 2-category, where each hom-category $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ is discrete with only identity 2-cells.

Furthermore, we define the following:

- $\Pi_G \subset \mathscr{F}_G$ denotes the subcategory where each morphism $\phi : \mathbf{n}^\alpha \to \mathbf{m}^\beta$ satisfies $|\phi^{-1}(j)| = 0, 1$ for all $j \neq 0$.
- $\Lambda_G, \Lambda_G^\perp, \Sigma_G \subset \Pi_G$ denotes the subcategory where each morphism is injective, surjective, and bijective respectively (in particular, $\Lambda_G \cap \Lambda_G^\perp = \Sigma_G$).
- $\mathscr{E}_G \subset \mathscr{F}_G$ denotes the subcategory where each morphism satisfies $\phi^{-1}(0) = \{0\}$.

6.1 2-Category $\mathscr{F}_G$-PsAlg

Definition 6.2 ($\mathscr{F}_G$-pseudoalgebra)**.** An $\mathscr{F}_G$-pseudoalgebra, is a pseudofunctor $\tilde{\mathscr{A}} : \mathscr{F}_G \to \mathbf{Cat}_G$, such that restricting to the subcategory $\Pi_G \subset \mathscr{F}_G$, $\tilde{\mathscr{A}}$ is a strict functor. Moreover, we require it to be reduced, namely $\tilde{\mathscr{A}}(0) = *$ (the terminal G-category), and each hom-functor

$$\tilde{\mathscr{A}}_{\mathbf{n}^\alpha, \mathbf{m}^\beta} : \mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \to \mathbf{Cat}_G(\tilde{\mathscr{A}}(\mathbf{n}^\alpha), \tilde{\mathscr{A}}(\mathbf{m}^\beta))$$

is a G-functor.

Let $\delta_i : \mathbf{n}^\alpha \to \mathbf{1}$ be the ith projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G, it generates a functor $\tilde{\mathscr{A}}\delta_i : \tilde{\mathscr{A}}(\mathbf{n}^\alpha) \to \tilde{\mathscr{A}}(\mathbf{1})$. Hence, by universality of product, all $\tilde{\mathscr{A}}\delta_i$ $(1 \leq i \leq n)$ induces a functor $\delta : \tilde{\mathscr{A}}(\mathbf{n}^\alpha) \to \tilde{\mathscr{A}}(\mathbf{1})^{\mathbf{n}^\alpha}$.

Definition 6.3 ($\mathscr{F}_{G,\mathrm{vs}}$ and $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra)**.** An $\mathscr{F}_G$-pseudoalgebra $\tilde{\mathscr{A}}$ is *very special* (or an $\mathscr{F}_{G,\mathrm{vs}}$-pseudoalgebra), if $\delta : \tilde{\mathscr{A}}(\mathbf{n}^\alpha) \to \tilde{\mathscr{A}}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathscr{F}_G$, and $\tilde{\mathscr{A}}$ is *strictly special* (or an $\mathscr{F}_{G,\mathbb{R}}$-pseudoalgebra), if all δ is in fact identity.

Definition 6.4 ($\mathscr{F}_G$-1-cell)**.** Given $\tilde{A}, \tilde{B}$ two $\mathscr{F}_G$-pseudoalgebras, a morphism (or $\mathscr{F}_G$-1-cell) between them is a pseudo-transformation $(\mathbb{F}, \zeta) : \tilde{\mathscr{A}} \to \tilde{\mathscr{B}}$, such that restricting to Π_G, it becomes an actual natural transformation. Moreover, it respects G-equivariance for every $g \in G$ and morphism $f \in \mathrm{Mor}(\mathscr{F}_G)$:

$$\zeta(g \circ f \circ g^{-1}) = g \circ \zeta(f) \circ g^{-1}$$

Definition 6.5 ($\mathscr{F}_G$-2-cell)**.** Given $\tilde{A}, \tilde{B}$ two $\mathscr{F}_G$-pseudoalgebras, and two $\mathscr{F}_G$-1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{A} \to \tilde{\mathscr{B}}$, an $\mathscr{F}_G$-2-cell is a transformation $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$, such that for any morphism $f : \mathbf{n}^\alpha \to \mathbf{m}^\beta$ in $\mathscr{F}_G$, the following two pasting diagrams are equal:

$$
\begin{tikzcd}
\tilde{\mathscr{A}}(\mathbf{n}^\alpha) \arrow[r, bend left=40, "\mathbb{F}_{\mathbf{n}^\alpha}"] \arrow[r, bend right=40, "\mathbb{G}_{\mathbf{n}^\alpha}"'] \arrow[r, phantom, "\Downarrow\lambda"] \arrow[d, "\tilde{\mathscr{A}}f"'] & \tilde{\mathscr{B}}(\mathbf{n}^\alpha) \arrow[d, "\tilde{\mathscr{B}}f"] \arrow[dl, phantom, "\swArrow \xi f"] \\
\tilde{A}(\mathbf{m}^\beta) \arrow[r, "\mathbb{G}_{\mathbf{m}^\beta}"'] & \tilde{B}(\mathbf{m}^\beta)
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
\tilde{\mathscr{A}}(\mathbf{n}^\alpha) \arrow[r, "\mathbb{F}_{\mathbf{n}^\alpha}"] \arrow[d, "\tilde{\mathscr{A}}f"'] & \tilde{\mathscr{B}}(\mathbf{n}^\alpha) \arrow[d, "\tilde{\mathscr{B}}f"] \arrow[dl, phantom, "\swArrow \zeta f"] \\
\tilde{A}(\mathbf{m}^\beta) \arrow[r, bend left=40, "\mathbb{F}_{\mathbf{m}^\beta}"] \arrow[r, bend right=40, "\mathbb{G}_{\mathbf{m}^\beta}"'] \arrow[r, phantom, "\Downarrow\lambda"] & \tilde{B}(\mathbf{m}^\beta)
\end{tikzcd}
$$

7 Construction of $\mathbb{R}_G$ (ND)

7.1 Construction of Objects

7.2 Extension to 2-functor

Reference Not Done (find their corresponding journal, if not the arxiv)

References

[1] K. Bangs, S. Binegar, Y. Kim, K. Ormsby, A. M. Osorno, D. Tamas-Parris and L. Xu. Biased Permutative Equivariant Categories. Homology, Homotopy and Applications, vol. 23. 2021. https://arxiv.org/abs/1907.00933

[2] P. Bonventre, L. A. Pereira. Genuine Equivariant Operads. Available as ArXiv:0717.02226, 2020.

[3] B. J. Guillou and J. P. May. Equivariant Iterated Loop Space Theory and Permutative G-Categories. Avilable as ArXiv:1207.3459, 2017.

[4] B. J. Guillou, J. P. May, M. Merling, A. M. Osorno. A Symmetric Monoidal and Equivariant Segal Infinite Loop Space Machine. Available as AXiv:1711.09183, 2018.

[5] B. J. Guillou, J. P. May, M. Merling, A. M. Osorno. Symmetric Monoidal G-Categories and their Strictification. Available as ArXiv:1809.03017, 2019.

[6] B. J. Guillou, J. P. May, M. Merling, A. M. Osorno. Multiplicative Equivariant K-Theory and the Barratt-Priddy-Quillen Theorem. Available as ArXiv:2102.13246, 2021.

[7] N. Johnson, D. Yau. 2-Dimensional Categories. Available as ArXiv:2002.06055, 2020.

[8] H. J. Kong, J. P. May, F. Zou. Orbital Presheaves in Equivariant Infinite Loop Space Theory. Available as ArXiv:2607.12141, 2026.

[9] Jiasen Liu, J. P. May, Kyle I. Roke, Hongyi Zhang and Keming Zhou. What are Symmetric Monoidal Categories? Available as ArXiv:2607.13912, 2026.

[10] J. P. May. Operads, Algebras and Modules. In *Operads: Proceedings of Renaissance Conferences (Hartford, CT/Luminy, 1995)*, volume 202 of *Contemp. Math.*, pages 15-31. Amer. Math. Soc., Providence, RI, 1997.

[11] S. Mac Lane. Categories for the Working Mathematician. 2nd ed. Graduate Texts in Mathematics 5. New York, NY: Springer, 1998.

[12] J. P. May, M. Merling, and A. M. Osorno. Equivariant Infinite Loop Space Theory, the Space Level Theory. Available as ArXiv:1704.03413, 2025.

[13] E. Riehl. Category Theory in Context. Aurora: Modern Math Originals, Dover Publications, 2017.